

Tyler McCormack

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Education

Northeastern University- Boston, MA

GPA: 3.90/4

April 2026

- PhD Candidate | Civil Engineering | Water, Environmental, and Coastal Systems
Dissertation: Advances in Remote Measurements of Nearshore Hydrodynamics

- Masters of Science in Civil and Environmental Engineering

Fall 2021

Rensselaer Polytechnic Institute (RPI) – Troy, NY

GPA: 3.65/4

- Bachelor of Science in Civil Engineering | Geotechnical Concentration | Sustainability Studies Minor

May 2020

Nanyang Technological University | Singapore

Spring 2019

- Study abroad semester

Postdoctoral Appointments

University of Grenoble Alpes | Institute of Environmental Geosciences | Grenoble, France

Sept 2026- March 2027

- Quantifying the effects of nearshore waves on the Surface Water and Ocean Topography (SWOT) satellite mission by combining SWOT pixel cloud with in-situ LiDAR measurements and studying their relationships with machine learning.
- Fulbright US Scholar Fellowship

Research Interests

- Applying remote sensing to quantify and characterize environmental processes to improve understanding of the natural world
- Remote sensing algorithm development
- Advancing remote observation of coastal processes, including wave mechanics, currents, and sediment transport
- Linking easily observable parameters to difficult-to-measure processes to expand the utility of remote sensing

Journal Publications

- (1) Beckman, J., Hopkins, J., Beighley, R., **McCormack, T.** 2026. Camden, Maine Coastal Storm Impacts and Mitigation: Framework and Case Study. *In preparation.*
- (2) Beckman, J., Hopkins, J., **McCormack, T.**, Wise, L., Cira, G. 2026. Quantifying the wave attenuation capacity of a vegetated floating canopy. *In preparation.*
- (3) **McCormack, T.**, Beighley, R., 2026. Applications, Limitations, and Insights for SWOT in Wave Dominated Nearshore Environments. *Geophysical Research Letters. In review*
- (4) Mansur, M., Chen, Q., Hopkins, J., Salatin, R., **McCormack, T.**, Dickhudt, P., 2025. Field Observations and Modeling of Wave Attenuation over an Artificial Vegetation Patch in the Swash Zone. *Coastal Engineering. In review*
- (5) **McCormack, T.**, Hopkins, J., Raubenheimer, B., Elgar, S., Brodie, K.L., 2025. Remote sensing of wave-orbital velocities in the surfzone. *Coastal Engineering* 195, 104631. doi.org/10.1016/j.coastaleng.2024.104631

Conference Presentations

- (1) McCormack, T., Beighley, E., SWOT in the nearshore: Extending HR Pixel cloud product to the surfzone for hyperlocal wave measurements, *Surface Water and Ocean Topography (SWOT) Science Team Meeting*, Arcachon, France, 14-17 Oct 2025
- (2) McCormack, T., Hopkins, J., Predicting instantaneous subsurface velocity using surface velocity, *International Conference on Coastal Engineering*, Rome, Italy, 8-14 Sept, 2024, Abstract #: 538
- (3) McCormack, T., Hopkins, J., Relating Wave Geometry and Surface Dynamics to Subsurface Velocities, *International Conference on Coastal Engineering*, Sydney, Australia, 4-9 Dec 2022, Abstract #: 1320
- (4) McCormack, T., Sherman, A., Watkins, L., Obando, F., and Hopkins, J.: Evaluation of Delft3d Microplastic Model of the Mid-Atlantic New Jersey Coast Using Summer and Winter Field Measurements, EGU General Assembly 2022, Vienna, Austria, 23–27 May 2022, <https://doi.org/10.5194/egusphere-egu22-6655>

- (5) McCormack, T., Hopkins, J. Combining acoustic and optical measurements to understand the relationship between internal and external vortices of breaking waves. *Young Coastal Scientists and Engineers Conference- Americas*, Myrtle Beach, SC, October 2021.

Field Research

Harbor wave propagation and flooding studies | Boston, MA & Camden, ME *Spring 2023-Fall 2024*

Lead coastal field deployments by assembling, deploying, and maintaining water level sensors and Sofar Spotter buoys with Smart Moorings to collect high-resolution wave and water level data. Data were used for hydrodynamic model forcing and validation as part of studies on wave propagation, ship wake impacts, and flooding in urban harbors.

Design and Fabrication of Custom Surveying Jetski | Boston, MA *Fall 2023*

Designed and implemented a custom hydrographic survey platform integrating GNSS, IMU, and sonar for a jetski capable of cm-level bathymetric mapping in water as shallow as 50 cm.

Subsurface Profiles And Multispectral-drone EXperiment (SPAM-EX) | USACE FRF Duck, NC *Fall 2022*

Investigated wave breaking dynamics using colocated ADCP measurements and UAV-based LiDAR to relate subsurface orbital velocities to evolving free-surface wave geometry.

DURING Nearshore Events eXperiments (DUNEX) | USACE Field Research Facility (FRF) Duck, NC *Fall 2021*

Supported experiment to measure wave attenuation and morphological impacts of an artificial seagrass patch in the swash zone by designing, implementing, and maintaining a system of ten cameras in order to quantify flow velocities using particle image velocimetry (PIV).

Assessing the validity of velocity measurements with an GoPro camera | Newport, OR *Summer 2021*

Designed, coordinated, and executed a series of field experiments at South Beach State Park where colocated acoustic doppler velocimeter (ADV) and a GoPro camera measured sub-surface (ADV) and surface (GoPro) velocities to test the validity of velocity measurements measured with the GoPro camera and particle image velocimetry (PIV) techniques

Laboratory Research

Effect of vegetated artificial floating island on energy attenuation | Oregon State University *Summer 2022*

Built a single full scale physical model of an artificial floating island. Collected and analyzed data to determine how different vegetation configurations affect the efficacy of energy attenuation in various wave conditions wave flume.

Physical modeling of artificial floating island network | Oregon State University *Summer 2021*

Tested a 3/5 scale model of an artificial floating wetland, dubbed the Emerald Tutu, in various wave climates in the O.H. Hinsdale Directional Wave Basin to find the friction factor for future numerical models.

GeoWall Design Competition | Design and Build Team | Rensselaer Polytechnic Institute (RPI) *Fall 2019- Spring 2020*

Designed a model mechanically stabilized earth (MSE) wall to compete in the 2020 GeoCongress GeoWall Design Competition. The team placed first at the national competition at GeoCongress 2020.

Research Assistant | Boundary Control of Smart Water Management Systems | RPI *Spring 2020*

Led a team designing a laboratory flume to study sediment dynamics upstream of a hydroelectric dam and produced a report guiding future instrumentation.

Research Assistant | Bluff Body Vortex Shedding | Rensselaer Polytechnic Institute *Fall 2018*

Measured the effects of upstream free stream turbulence on the free shear layer of bluff bodies and performed vortex shedding analysis using Particle Image Velocimetry and MATLAB.

Modeling Research

Identifying microplastic convergence zones on with Mid-Atlantic coast with Delft3D and field data *Fall 2020*

Identified Mid-Atlantic microplastic convergence zones using a coupled Delft3D hydrodynamic-wave and particle tracking model and substantiated with field observations along the New Jersey coast.

Research Assistant | Near Field Tsunami Evacuation | Oregon State University *Summer 2019*
Studied the efficacy of community leaders in rapid onset near field tsunami evacuations using a multi-modal agent-based model. The framework integrated transportation networks, population distribution and inundation data to accurately portray the community, with plans to eventually implement the findings into evacuation protocol.

Coastal Protection Design using Computational Fluid Dynamics Simulations | RPI *Fall 2017*
Led integration of complex systems in an environmentally conscious seawall design and conducted CFD simulations (Flow-3D) of coastal wave conditions.

Educational Research

Research Assistant | Rensselaer Polytechnic Institute *Fall 2019- Summer 2020*
Collaborated with universities to develop a mixed reality platform enabling students to conduct pseudo fieldwork with real data. Coordinated between universities and implemented platform at universities nationally

Mentorship

Undergraduate Research Mentor | Northeastern University

- Sea level rise modeling with Delft3D *Spring 2023*
- Microplastic modeling with Delft3D *Fall 2022*
- Identifying wave breaking with computer vision *Fall 2021*
- Particle image velocimetry sensitivity study *Spring 2021*

Teaching

Teaching Assistant | Coastal Dynamics and Design | Northeastern University *Spring 2026*
Held weekly office hours to support student learning and graded assignments for a class of 20 students

Teaching Assistant | Fluid Mechanics | Northeastern University *Fall 2022*
In charge of leading the lab and recitation components of the course. Gave several lectures as well as hosted office hours regularly attended by the students.

Teaching Assistant | Fluid Mechanics for Civil Engineers | Rensselaer Polytechnic Institute *Fall 2019*
Prepared and supervised laboratory sessions, graded coursework and exams, and provided student support as needed.

Honors

Northeastern Civil and Environmental Engineering PhD Leadership Award *Spring 2025*
NSF NHERI RAPID Graduate Scholar *Fall 2022*

Awarded to two graduate students per year and funds student-led research proposals that allow the student to design and orchestrate a field experiment using the RAPID Facility equipment.

The Pickett T. Simpson Prize *Spring 2020*
Awarded to an undergraduate who demonstrated strong ability and shows promise for a career in engineering.

Dean's Honor Lis and Cum Laude Graduation Honors | Rensselaer Polytechnic Institute *Spring 2020*

National Civil Engineering Honor Society | Executive Board Member | Marshall *2018-2019*

Rensselaer Leadership Award *2016-2020*

Skills

MATLAB	Signal Processing	Particle Image Velocimetry (PIV)
Certified drone pilot	Hydrodynamic Modeling, Delft3D	QGIS
LiDAR data collection and processing	High Performance Computing	Python

Courses

Coastal Dynamics and Design	Coastal and Hydraulic Modeling	Coastal and Nearshore Hydrodynamics
Remote Sensing	Sediment Transport Applications	Environmental Fluid Mechanics
Mathematical Methods and Modeling	Time Series/ Geospatial Data Sci.	Finite Element Analysis (FEA)
Engineering Dynamics	Strength of Materials	Thermodynamics